

1. List 5 common BIOS or POST error messages you might see when starting a PC and write what hardware or configuration issue each one usually points to.

BIOS/POST Error Message	Hardware or Configuration Issue
1. CMOS Checksum Error	CMOS settings are corrupted or the CMOS battery is weak and needs replacement.
2. Keyboard Error or No Keyboard Detected	Keyboard is disconnected, faulty, or the USB/PS/2 port is not working.
3. No Boot Device Found	Hard drive/SSD is not detected, boot order is incorrect, or the operating system is missing.
4. Memory Test Failed	RAM is faulty, not seated properly, or incompatible with the motherboard.
5. CPU Fan Error	CPU cooling fan is disconnected, faulty, or not spinning properly, which may cause overheating.

2. Given this scenario: On boot, a PC beeps 3 times in a repeating pattern and does not display anything on the screen. Research and write what this POST beep code typically means for an AMI BIOS, and suggest two troubleshooting steps.

AMI BIOS POST Beep Code Scenario

Scenario: A PC beeps **3 short beeps repeatedly** during startup and does not display anything on the screen.

Meaning (AMI BIOS):

For a traditional AMI BIOS, **3 short beeps** usually indicate a **Base 64 KB RAM failure (memory read/write test error)**. This means the BIOS detected a problem with the system memory (RAM) during the Power-On Self-Test

Two Troubleshooting Steps

1. **Reseat the RAM modules**
 - Turn off the PC and unplug the power.
 - Remove the RAM sticks, clean the contacts if necessary, and firmly reinstall them in the correct slots.
2. **Test the RAM**
 - Boot the PC with **one RAM stick at a time** and try different memory slots.
 - If one module or slot consistently causes the beep code, replace the faulty RAM or inspect the motherboard memory slot.

3. Access the BIOS setup utility on any available PC or virtual machine, take a screenshot of the main BIOS screen, and identify the motherboard manufacturer and BIOS version.

- Restart your PC or virtual machine.
- Press the BIOS key during startup (commonly **Delete**, **F2**, **F10**, or **Esc**, depending on the manufacturer).
- Open the **Main** or **System Information** page in the BIOS.
- Take a screenshot or photo of the BIOS screen.
 - On a physical PC, you can use your phone to take a clear picture.
 - On a virtual machine, use the VM's screenshot feature or the host operating system's screenshot tool.
- Record the following information from the BIOS screen:
 - **Motherboard Manufacturer:** _____
 - **BIOS Version:** _____

Item	Example Value
Motherboard Manufacturer	ASUS
BIOS Version	3202

4. You are given a screenshot of a POST error message that says 'CMOS checksum error – Defaults loaded.' Explain what this means and describe how you would resolve this issue in a real system. Hint: Think about the CMOS battery and BIOS settings reset.

CMOS Checksum Error – Defaults Loaded

What it means:

The message "**CMOS checksum error – Defaults loaded**" means that the BIOS detected that the CMOS memory settings are corrupted or do not match the expected checksum. As a result, the BIOS automatically loaded its default settings. This commonly happens because the **CMOS battery is weak or dead**, the BIOS settings were reset, or the CMOS memory became corrupted.

How to Resolve the Issue

1. **Enter the BIOS Setup**
 - Restart the computer and press the BIOS key (such as **Delete** or **F2**) to enter the BIOS setup.
2. **Check and Reconfigure BIOS Settings**
 - Verify the **date and time**.
 - Restore any required settings such as the correct **boot order** or storage mode.
3. **Save and Restart**
 - Save the BIOS settings and reboot the computer.
 - If the error does not return, the issue may have been a temporary BIOS reset.

4. **Replace the CMOS Battery**

- If the error appears again after restarting or after the PC has been powered off, replace the **CR2032 CMOS battery** on the motherboard with a new one.

5. **Update BIOS (if needed)**

- If the problem continues even after replacing the battery and resetting the settings, update the BIOS to the latest version recommended by the motherboard manufacturer.

6. **Conclusion**

7. A **CMOS checksum error** usually indicates that the BIOS settings have been lost or corrupted, most often due to a **weak or dead CMOS battery**. Replacing the battery and reconfiguring the BIOS settings typically resolves the problem.